

Saurabh Surashe

MTS III at Mavenir Systems

+91-7276561155 || surashesaurabh21@gmail.com || *LinkedIn*

PROFESSIONAL SUMMARY

C++ and Linux Systems Software Engineer with **4+ years of experience** developing high-performance, latency-sensitive networking software for carrier-grade wireless communication systems.

Experienced in Layer-2 scheduler architecture, protocol stack development, Linux systems programming, concurrent software, performance optimization and production debugging.

Designed networking software supporting **NB-IoT NTN**, delivering **production features, protocol enhancements, performance optimizations and customer fixes**.

Passionate about Networking Infrastructure, Distributed Systems, High Performance Computing, AI Infrastructure and Modern C++ software development.

CORE COMPETENCIES

Programming Languages: C++17, STL, C, Python, Java

Linux & Systems: Linux, POSIX, pthreads, Shared Memory IPC, Multithreading, GDB, Intel VTune

Networking & Protocols: TCP/IP, UDP, NB-IoT, LTE, Non-Terrestrial Networks (NTN), MAC, RRC, S1AP, Wireshark, ASN.1

Software Engineering: Concurrent Programming, Distributed Systems, Performance Optimization, Low-Latency Systems, Carrier-Grade Software

AI / Machine Learning: Sentence Transformers, Semantic Search, Embeddings, TensorFlow, Keras

Development Tools: Git, Jenkins, Jira

PROFESSIONAL EXPERIENCE

Mavenir Systems Pvt. Ltd.

July 2022 – Present

Member of Technical Staff I → Member of Technical Staff III

Bengaluru, India

- Designed and developed high-performance Layer-2 and Layer-3 networking software in C/C++ for a carrier-grade NB-IoT wireless communication platform supporting both terrestrial and Non-Terrestrial Networks (NTN).
- Redesigned the latency-sensitive MAC scheduler resource allocation workflow, improving uplink throughput by approximately **40%** while preserving deterministic scheduling deadlines for real-time deployments.
- Architected key Layer-2 scheduler enhancements for **3GPP Release-17 Non-Terrestrial Networks (NTN)**, implementing Timing Advance computation, Doppler compensation, dynamic satellite ephemeris management and interlaced scheduling for GEO and LEO satellite communication.
- Designed configurable **Link Adaptation** algorithms leveraging radio quality measurements and HARQ feedback to maximize throughput while maintaining communication reliability under varying channel conditions.
- Architected a low-latency **Shared Memory IPC** framework enabling runtime propagation of satellite ephemeris information across distributed Layer-2 software components without service interruption.
- Implemented more than **60 production KPIs** providing detailed visibility into scheduler behaviour, protocol execution, resource utilization and overall software health, substantially improving runtime observability and customer debugging.
- Profiled latency-sensitive software using **Intel VTune**, Linux perf and GDB to identify CPU bottlenecks, optimize execution paths and improve runtime performance.
- Collaborated closely with Layer-1, Layer-2, Layer-3, Platform, QA, DevOps and Customer Engineering teams throughout architecture, implementation, validation and commercial deployment.
- Mentored two junior engineers through design discussions, debugging sessions, code reviews and knowledge sharing while promoting engineering best practices.

PROJECTS

AI-Powered Job Discovery & Matching Platform | *Python, FastAPI, SQL, Sentence Transformers, SMTP*

- Developed a modular Python platform automating job discovery and semantic matching using the **all-MiniLM-L6-v2** transformer model to calculate resume-to-job description similarity scores.
- Built asynchronous scraping pipelines using **asyncio** to concurrently fetch and parse opportunities from major ATS engines including Workday, Greenhouse, Eightfold, and SmartRecruiters.
- Implemented highly optimized, in-memory semantic similarity search utilizing NumPy vector calculations, achieving sub-200ms scoring latency without database-side vector extensions.
- Designed clean REST APIs with FastAPI to support resume parsing, job indexing, user subscription profiles, and score-threshold configurations.
- Integrated an automated notification system sending formatted HTML job digests via SMTP for jobs matching custom score thresholds, deployed via serverless pipelines and background tasks.

ACHIEVEMENTS

Mavenir Spotlight Award (Received Multiple Times) for delivering high-impact production features, technical excellence and successful customer software releases.

Co-authored: **“Privacy and Security Comparison of Web Browsers: A Review”**, AINA 2022 (*SpringerLink*)

Co-authored: **“Enhancing Cyber Security: Malicious web page detection using Machine and Deep Learning”**, AINA 2025 (*SpringerLink*)

EDUCATION

National Institute of Technology Surathkal

Master of Technology in Computational Data Science

Karnataka

June, 2020 – May, 2022

MIT College of Engineering Pune

Bachelor of Engineering in Mechanical Engineering

Maharashtra

June, 2015 – May, 2019